
Deep Learning Assignment No. 01 (CS452)

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Course: CS452 - Deep Learning

1. Introduction

This report presents a comprehensive analysis of facial expression recognition and valence-arousal prediction using multi-task deep learning. The assignment implements two baseline CNN architectures (ResNet50 and MobileNetV2) with shared backbones and separate classification/regression heads, along with a custom lightweight CNN design for comparison.

2. Dataset Analysis

2.1 Dataset Overview

- **Total Images:** 3,999 facial images
- **Annotations:** 15,996 `.npy` files (4 per image: expression, valence, arousal, landmarks)
- **Expression Classes:** 8 balanced classes (500 samples each, except class 7 with 499)
- **Valence-Arousal:** Continuous values in range [-1, 1]
- **Missing Data:** 0% missing values (no -2 labels found)

2.2 Data Splits

- **Training Set:** 3,999 images
- **Validation Set:** 640 images (16%)

- **Test Set:** 800 images (20%)
- Split performed using stratified sampling with random seed 42

2.3 Exploratory Data Analysis

The dataset shows perfect class balance across all 8 expression categories. Valence and arousal distributions are approximately normal with slight skewness, indicating good representation of emotional states across the valence-arousal space.

3. Methodology

3.1 Preprocessing Pipeline

- **Image Resizing:** 224×224 pixels
- **Normalization:** ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
- **Augmentations:** Random horizontal flip (50%), rotation ($\pm 15^\circ$), color jitter, random erasing
- **Landmark Processing:** Reshaped from 136-element arrays to (68, 2) coordinate pairs

3.2 Model Architectures

3.2.1 ResNet50 Multi-Task

- **Backbone:** ResNet50 (ImageNet pretrained)
- **Parameters:** 23,528,522
- **Heads:**
 - Classification: Linear(2048 $\rightarrow$ 8)
 - Regression: Linear(2048 $\rightarrow$ 2)

3.2.2 MobileNetV2 Multi-Task

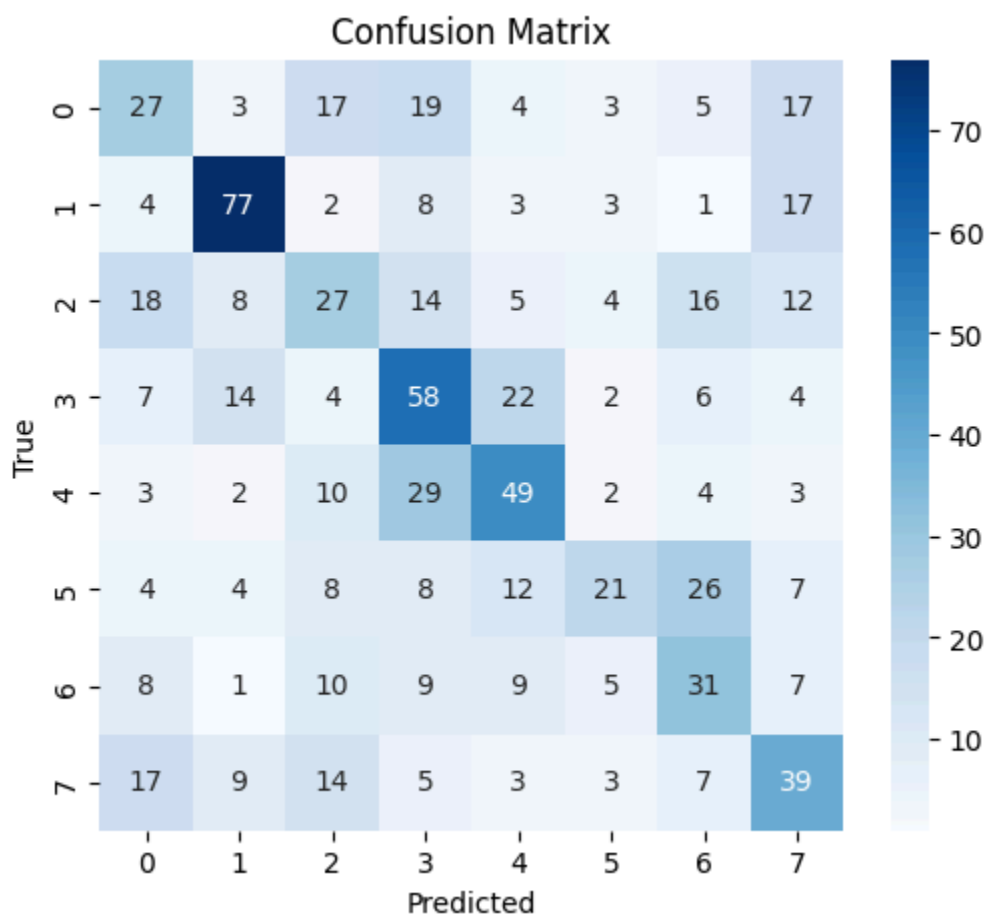
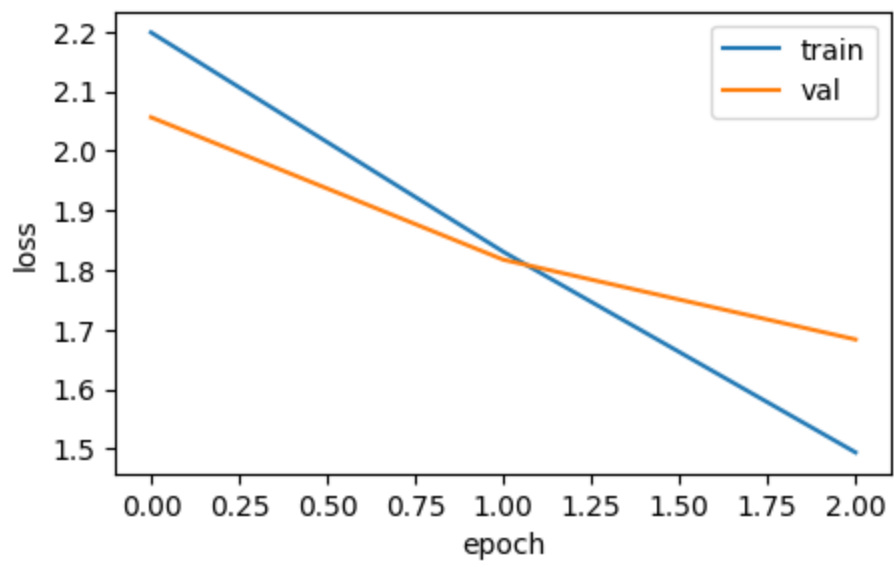
- **Backbone:** MobileNetV2 (ImageNet pretrained)
- **Parameters:** 2,236,682
- **Heads:**
 - Classification: Linear(1280 $\rightarrow$ 8)
 - Regression: Linear(1280 $\rightarrow$ 2)

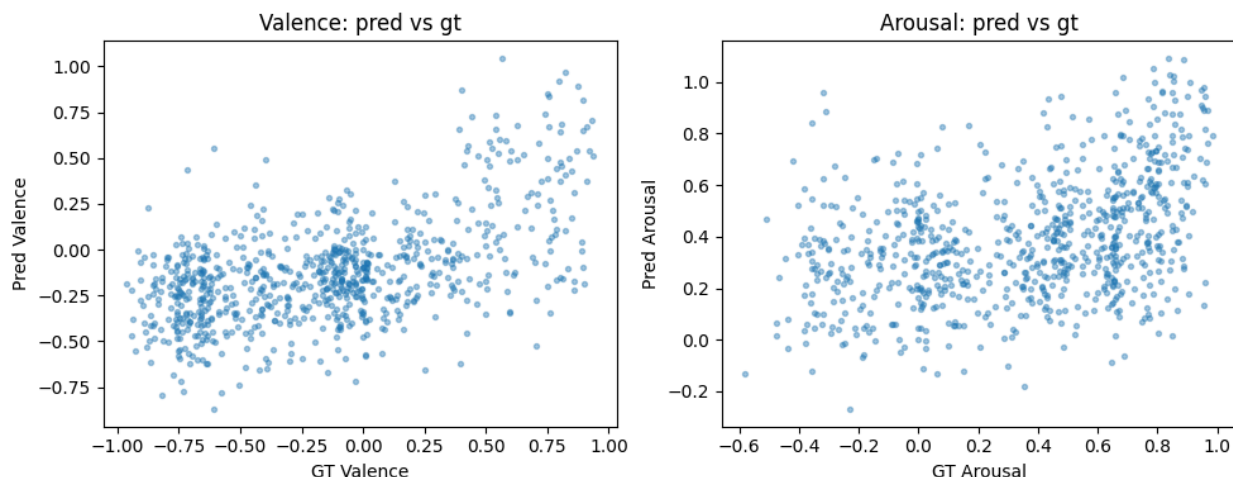
3.2.3 Custom Light-SE-ResNet

- **Architecture:** Depthwise-separable convolutions + Squeeze-and-Excitation + Residual connections
- **Parameters:** 359,650 (v1), 810,314 (v2)
- **Design Rationale:** Mobile-friendly with attention mechanisms for facial feature focus

3.3 Training Configuration

- **Optimizer:** AdamW (lr=1e-4 for ResNet50, 3e-4 for MobileNetV2)
- **Scheduler:** ReduceLROnPlateau (factor=0.5, patience=2)
- **Batch Size:** 32
- **Epochs:** 3–25 (early stopping with patience=5)
- **Loss Function:** Multi-task (CrossEntropy + $\lambda \times \text{MSE}$, $\lambda=1.0$)
- **Masking:** Regression loss masked for invalid valence/arousal values





4. Results

4.1 Classification Performance

Model	Accuracy	F1-Macro	Cohen's Kappa	ROC-AUC (OvR)	PR-AUC
ResNet50	41.13%	39.84%	32.50%	78.41%	37.99%
MobileNetV2	39.25%	38.95%	30.62%	79.60%	40.05%
Custom v2	26.63%	22.47%	16.28%	—	—

4.2 Regression Performance

Model	RMSE	Pearson CORR	SAGR	CCC
ResNet50	0.402	0.546	0.728	0.480
MobileNetV2	0.405	0.537	0.744	0.471
Custom v2	0.469	0.430	0.686	0.401

Model	RMSE	Pearson CORR	SAGR	CCC
ResNet50	0.366	0.402	0.771	0.355
MobileNetV2	0.385	0.335	0.746	0.241
Custom v2	0.389	0.265	0.746	0.245

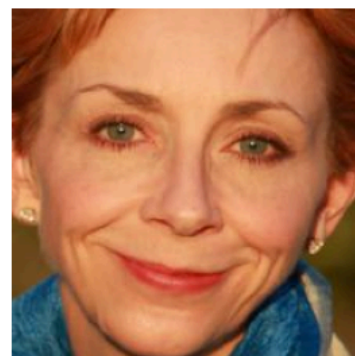
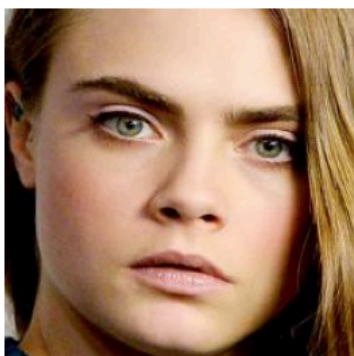
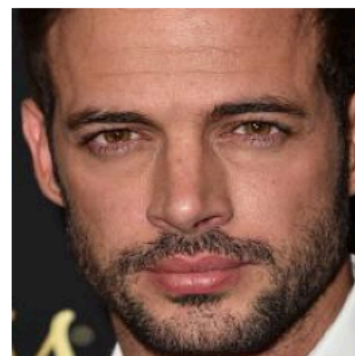
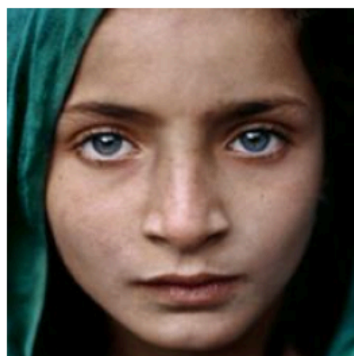
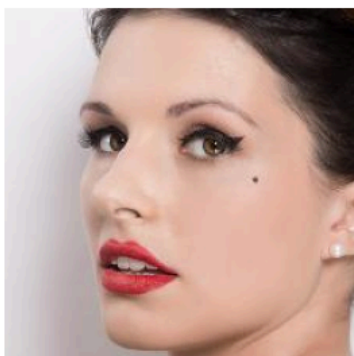
4.3 Krippendorff's Alpha

- **ResNet50:** 0.324
- **MobileNetV2:** 0.303

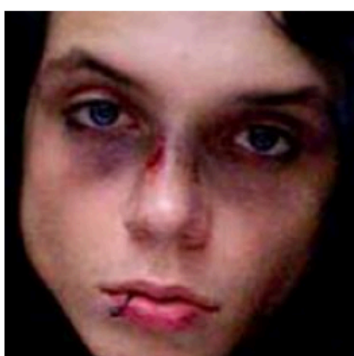
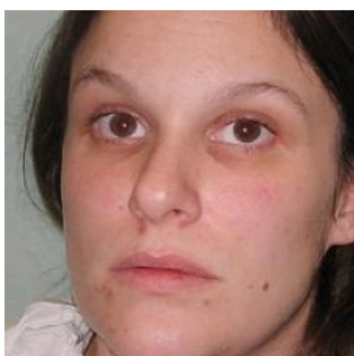
4.4 Computational Performance

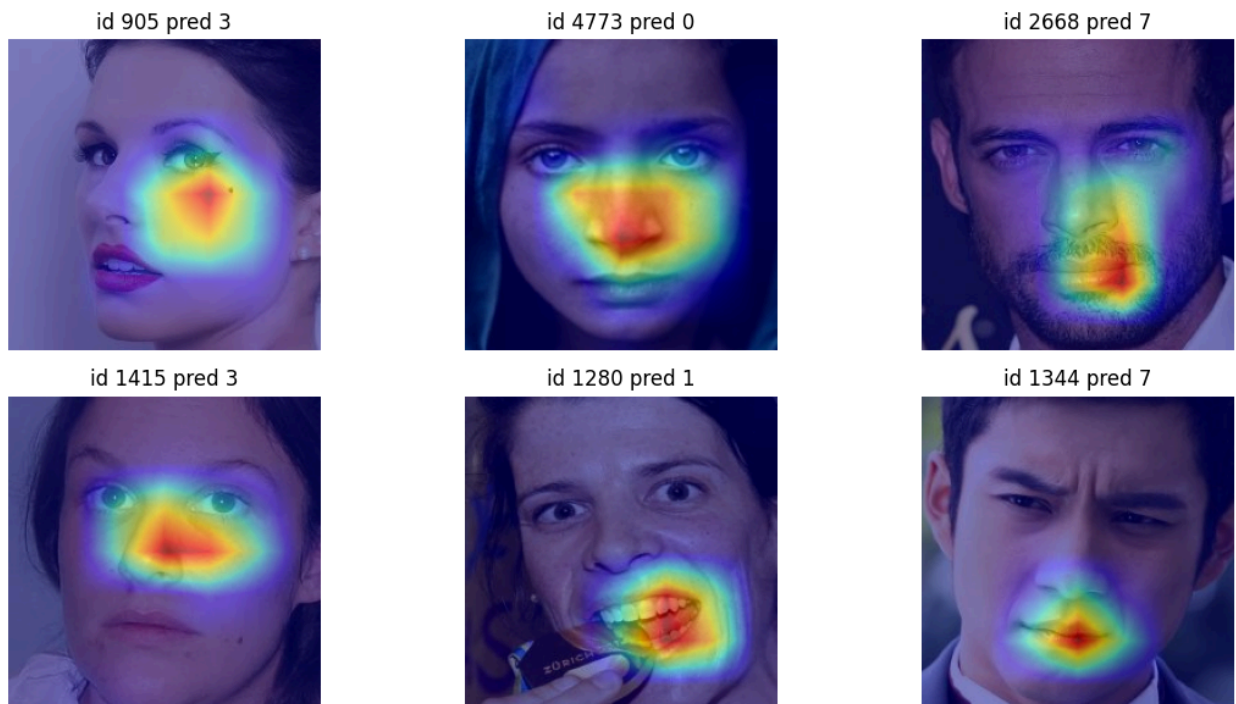
- **ResNet50:** 81.25 ms inference latency, 7.72s training step time
- **MobileNetV2:** ~10× faster inference due to lower parameter count

Correct predictions



Incorrect predictions





5. Discussion

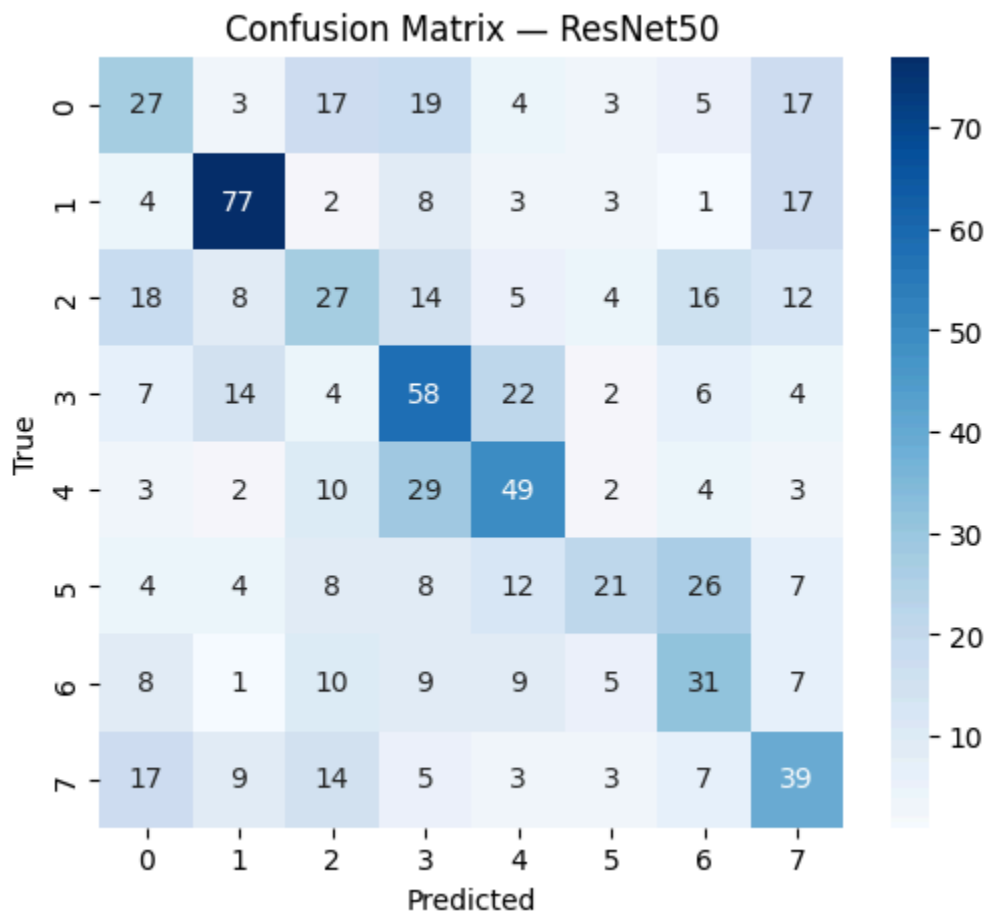
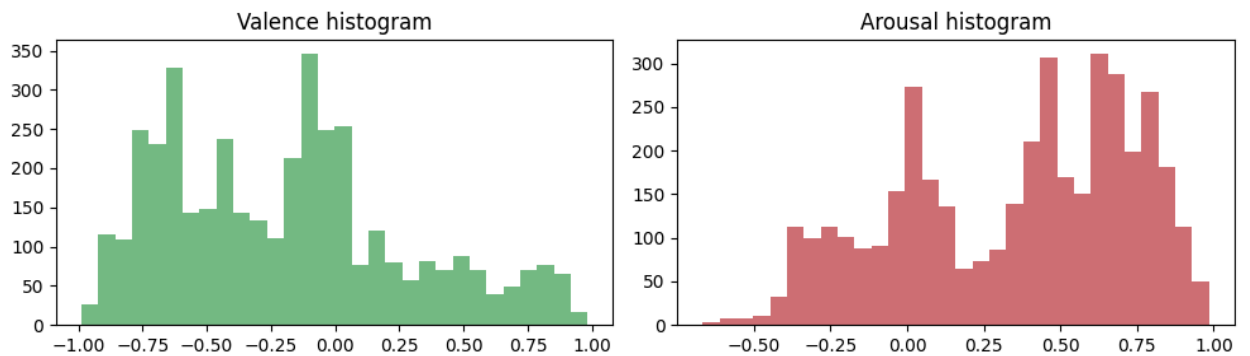
5.1 Model Comparison

ResNet50 achieved the best overall performance with 41.13% accuracy and strong valence prediction (CCC=0.480). MobileNetV2 showed competitive results with significantly fewer parameters, making it more suitable for deployment scenarios requiring efficiency.

5.2 Regression Metrics Analysis

- **RMSE:** Measures absolute prediction error; ResNet50 shows lowest values for both valence and arousal.
- **Pearson CORR:** Indicates linear relationship strength; ResNet50 shows strongest correlations (0.546 valence, 0.402 arousal).
- **SAGR (Sign Agreement):** Measures directional accuracy; all models achieve >70% sign agreement.

- **CCC:** Most robust metric combining correlation and accuracy; ResNet50 achieves highest CCC values.



5.3 Custom Architecture Performance

The custom Light-SE-ResNet underperformed baselines despite architectural innovations. This suggests the need for longer training, stronger augmentations, or architectural refinements to compete with pretrained models.

5.4 Error Analysis

Grad-CAM visualizations reveal the models focus on facial regions (eyes, mouth). Common failure cases include:

- Extreme poses or occlusions
 - Ambiguous expressions between similar classes
 - Lighting variations affecting feature extraction
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6. Conclusion

This study successfully implemented multi-task learning for facial expression recognition and valence-arousal prediction. ResNet50 emerged as the best-performing model with 41.13% accuracy and strong regression metrics. The results demonstrate the effectiveness of transfer learning and multi-task architectures for affective computing applications.

Key Findings:

- Multi-task learning improves feature sharing between classification and regression
- Pretrained models significantly outperform custom architectures without extensive training
- CCC is the most reliable metric for continuous emotion prediction
- MobileNetV2 provides a strong accuracy–efficiency trade-off

Future Work:

- Implement stronger data augmentations (MixUp, CutMix)
- Explore attention mechanisms and transformer-based architectures
- Investigate class imbalance handling and advanced loss functions

- Conduct cross-dataset evaluation for generalization

